Received: 22 October 2022 | Revised: 24 July 2023 | Accepted: 5 August 2023

DOI: 10.1002/nav.22148

RESEARCH ARTICLE

WILEY

Creating a merger option to strategically prevent a merger among suppliers or customers

Lezhen Wu[1] | **Dingwei Gu**[1] | **Zhiyong Yao**[1] | **Wen Zhou**[2]

[1]School of Management, Fudan University, Shanghai, China
[2]Faculty of Business and Economics, The University of Hong Kong, Hong Kong, China

Correspondence
Dingwei Gu, School of Management, Fudan University, Room 338, Siyuan Building, 670 Guoshun Road, Shanghai, China.
Email: dwgu@fudan.edu.cn

Funding information
National Natural Science Foundation of China, Grant/Award Numbers: 72003037, 71873036, 72192844, 72192845

Abstract

It is well-established in the literature that a horizontal merger in a supply chain is profitable (or beneficial to firms at another tier) if the merger synergy exceeds some threshold referred to as the profitable (or beneficial) threshold. Our paper goes one step further by finding that the profitable threshold is always lower than the beneficial threshold, which implies that a firm may have an incentive to prevent a horizontal merger between its suppliers or customers, but never wants to precipitate one. Moreover, we propose a strategy to achieve the preventive goal—a firm can develop an *instrumental* merger option with another rival firm, which can help prevent the *target* merger in one of two ways. First, if the synergy of the instrumental merger is high relative to the synergy of the target merger, the firm can carry out the instrumental merger—even an unprofitable one—preemptively to make the target merger unprofitable. Second, if the synergy of the target merger is not too high and the synergy of the instrumental merger is moderate, that is, the instrumental merger is profitable and will inflict severe harm on the firms in the target merger, the firm can reserve the instrumental merger option as a deterrent and the profitable target merger is deterred. An interesting implication of the deterrent role of the instrumental merger option is that when the firm has two candidate merger partners, it may choose the partner with a lower synergy for the deterrence purpose.

KEYWORDS

horizontal merger, preemptive merger, strategic deterrent, supply chain management

1 | INTRODUCTION

The previous literature studies horizontal mergers mainly in a single-tier model. There is a new and growing literature that extends horizontal mergers in a supply chain environment (Cho, 2014; Zhu et al., 2016; Fee & Thomas, 2004; Shahrur, 2005, etc.). This is a natural result of practical needs. Industries are connected through a network of input and output. A horizontal merger will affect not only its rivals but also its suppliers and customers in the supply chain, the latter of which can be demonstrated by suppliers and customers' responses in many cases of horizontal mergers. For example, when United Technologies Corp. announced to buy Rockwell Collins Inc., Boeing and Airbus, the two largest aircraft makers, were both angry about the merger between two of their major suppliers. Boeing even threatened to lobby regulators to stop the deal, but in vain.[1]

Our paper focuses on two relevant questions that remain unexplored. First, what are the firms' attitudes toward a merger at another tier, either to precipitate or prevent it? Intuitively, firms would have an incentive to precipitate a merger if the merger is unprofitable but beneficial to them, or to prevent a merger if the merger is profitable but harmful to them.[2]

[1]"United Technologies merges with Rockwell Collins," The Economist, Sept. 9, 2017, https://www.economist.com/business/2017/09/09/united-technologies-merges-with-rockwell-collins.

[2]A merger is profitable if the post-merger profit is larger than the joint profits of pre-merger firms. A merger is beneficial to firms in other tiers of the supply chain if it increases the profit of firms in other tiers of the supply chain.

Second, can a firm achieve the goal of precipitating or preventing a merger at another tier by opening up a merger option?

To address these two questions, we develop a two-tier supply chain model similar to Cho (2014) where firms at the downstream tier take the input price as given and firms at each tier conduct Cournot competition, that is, successive Cournot, and a two-firm horizontal merger may generate a cost synergy in the form of a cost reduction. Consistent with the previous literature (Cho, 2014; Zhu et al., 2016), a merger has two opposite effects on the merging firms and firms at the other tiers along the supply chain—*concentration effect* and *efficiency effect*. A merger increases concentration, which tends to hurt all firms at the other tiers, and at the same time, the merger achieves a cost synergy, which tends to benefit them. Therefore, a merger is profitable if and only if the synergy is higher than some threshold (referred to as *profitable threshold*) such that the efficiency effect dominates. Similarly, a merger benefits its suppliers or customers if and only if the synergy is higher than some threshold (referred to as *beneficial threshold*). This provides one possible answer to the question that why Boeing was so angry at the merger between its suppliers—the merger increases the concentration at the upstream tier but does not achieve a large enough cost synergy.

Our first new finding is that firms will never bother to do anything to induce their suppliers or customers to merge, but may have an incentive to prevent their suppliers or customers from merging. This is a direct result of the comparison between the profitable threshold and the beneficial threshold. We find that the beneficial threshold is always higher than the profitable threshold, that is, it is more demanding for a merger to benefit firms at the other tiers. This comparison implies that it is possible for the merger to be profitable but harmful to their suppliers or customers (when the synergy falls between the two thresholds), but it is impossible for the merger to be unprofitable but beneficial to their suppliers or customers. Thus, we identify a preventive motive.

Given that firms may have a motive to prevent a merger at an up- or downstream tier under certain circumstances, a natural question is what strategy a firm can adopt to prevent the merger. It is generally rather difficult for a company to prevent another merger by relying on government intervention. Typically, the regulator scrutinizes a merger for the sake of consumers or social welfare instead of the interests of the merging firms' rivals, suppliers or customers, and thus lobbying is rarely effective. Besides, some countries even lack a mature antitrust law or institution.

Literature finds that mergers at one tier can affect the likelihood of mergers in another tier (Becker & Thomas, 2010; Bhattacharyya & Amrita, 2011). For one example, in personal computer industry, Hewlett-Packard and Compaq merged in 2001. Shortly after their merger, their suppliers, Seagate and Maxtor merged in 2006; Flextronics and Solectron merged in 2007.[3] For another example, after the media firms, Viacom and Paramount, Disney and ABC, Time Warner and Turner Broadcasting, carried out mergers since 1996, the record firms, Vivendi and Universal, AOL and Time Warner, Sony and BMG also carried out mergers sequentially. There is a strand of literature concerning the use of one merger for the strategic purpose of preventing or precipitating another merger. For example, Fridolfsson and Stennek (2005) find that one firm may acquire a target firm to prevent the target firm from being acquired by another rival firm; Gorton et al. (2009) find that one firm may acquire a target firm to become bigger in order to prevent itself from being acquired. On the other hand, Qiu and Zhou (2007) find that firms may conduct mergers strategically to trigger more mergers.

Inspired by the aforementioned literature and empirical evidence about the interaction between mergers at different tiers along the supply chain, we theoretically propose a strategy for firms to achieve the preventive goal: A firm can open up a merger option with another rival firm (referred to as *instrumental merger*) to prevent another harmful merger between its suppliers or customers (referred to as *target merger*). Relevant to our research question, we consider the parameter space where a target merger is profitable but harmful to firms at another tier, that is, target merger will be conducted and firms at another tier will be hurt in the absence of merger option. Following the literature, we assume that each pair of firms with a merger option decide whether to merge and the two pairs make decisions in turn.[4] Our model shows that the existence of merger option, either carried out or not, can help prevent the harmful merger at another tier if the synergies of the two mergers satisfy certain conditions.

First, if the synergy of instrumental merger is high enough relative to the synergy of target merger, then instrumental merger can discourage the target merger. Consistent with findings in Salant et al. (1983) and Qiu and Zhou (2007), the profitable threshold increases in the pre-merger profit margin. A merger with a high enough synergy will lower the overall marginal cost at that tier, and increase the profit margin and the profitable threshold of a merger at a different tier. That is, if the synergy of the instrumental merger is high enough, then carrying out the instrumental merger will make the target merger unprofitable. There may be a first-mover advantage in some cases, that is, each firm wants to merge as a first mover, and the second mover will not merge.

Second, if the synergy of target merger is not too high and the synergy of instrumental merger falls within some range, preserving the merger option of instrumental merger can deter target merger. That is, firms at another tier will refrain

[3]The Seagate and Maxtor are hard disk driver suppliers. The Flextronics and Solectron buy parts from hard disk drive suppliers, and are the customers of the Hewlett-Packard and Compaq.

[4]Since mergers are observable and irreversible, a sequential game is a more appropriate setup than a simultaneous game to model the strategic interaction of mergers. This is a common approach in the literature of mergers (Fauli-Oller, 2000; Nilssen & Sørgard, 1998; Qiu & Zhou, 2007).

from merging to avoid triggering the instrumental merger. The deterrent role of a merger option has several interesting managerial implications. First, a merger option, even not conducted, can have a strategic deterrent value, which equals the harm by another merger if the option is absent. Therefore, even if preserving a merger option is costly, it is worthwhile to maintain it as long as the maintenance cost is smaller than the deterrent value.[5] Second, for merger option to qualify as a deterrent to target merger, the synergy of instrumental merger must be neither too low nor too high. If the synergy is too low, its threat (of carrying out its own merger after target merger) is incredible; if the synergy is too high, carrying out instrumental merger can not hurt the firm at another tier too much. The latter explains a counterintuitive phenomenon identified in our model: A firm can be worse off when the synergy of its merger option becomes higher, because the deterrent effect disappears. When a firm has two acquisition targets with different synergies, it may be optimal to choose the one with a lower synergy for deterrence purpose. To enhance the credibility of the threat, firms can even start the negotiation process, which is costly and time-consuming.

The above two preventive methods—carrying out the merger or preserving the option—explain two seemingly confusing phenomena. When only the method of carrying out the merger works, firms will carry out an unprofitable merger.[6] When the merger is deterred by the merger option of firms at another tier, firms will withhold a profitable merger.

The rest of the paper is organized as follows: Section 2 reviews the related literature; Section 3 shows the model setups; Section 4 analyses the equilibrium to investigate the conditions under which a merger option can prevent a merger at another tier; Section 5 provides several implications for our main results; Section 6 concludes.

2 | RELATED LITERATURE

Our paper is built upon the work that studies horizontal mergers in a supply chain environment. There are a few papers that study the effects of a horizontal merger on the merging firms, suppliers, and customers. For theoretical work, Cho (2014) and Zhu et al. (2016) uncover conditions under which a merger is profitable in a supply chain and conditions under which it benefits the suppliers and customers. They consistently find that the cost saving of a horizontal merger benefits firms at other tiers along the supply chain while the concentration effect adversely affects them. For empirical work, Fee and Thomas (2004) and Shahrur (2005) use large samples of horizontal mergers to estimate effects of mergers on the merging firms and their customers and suppliers with the aim of uncovering the nature of gains from mergers in a supply chain. Shahrur (2005) finds that efficiency considerations are the primary motives of horizontal mergers and mergers that tend to benefit the merging firms tend to benefit firms at the other tiers. Based on the previous work, we model the mergers in such a way that cost synergy is the only non-strategic motive.

Our paper contributes to the literature in three ways.

First, though we do observe a practical need of firms to prevent a merger between suppliers or between customers, little has been done on this topic in the literature. Our paper fills the gap by offering a solution to this problem. The existing literature focuses on the preemptive motive of mergers within the same industry (Fridolfsson & Stennek, 2005; Gorton et al., 2009) and offers an explanation to the puzzle of unprofitable mergers being carried out: A firm may acquire a rival to preempt another rival from acquiring it. We extend the preemptive motive across tiers in a supply chain. We find that firms may merge preemptively to discourage a merger at the up- or downstream tier, or use a merger option to deter it. The former case gives another explanation to the puzzle.

Second, our paper contributes to the prior work on strategic motives of mergers. The traditional literature focuses on the non-strategic concerns of mergers. Motives of non-strategic mergers include a cost reduction (Gupta & Gerchak, 2002; Salant et al., 1983), a larger market power or bargaining power relative to a firm's suppliers or customers (Galbraith, 1952; Horn & Wolinsky, 1988; Lommerud et al., 2005; O'Brien & Shaffer, 2005), risk pooling (Cho & Wang, 2016; Xiao, 2019), and so forth. There is a growing literature that studies strategic mergers, that is, mergers conducted strategically to influence the behavior of other firms—either rival firms, suppliers, or customers. For example, Inderst and Wey (2003) show that retailers may consolidate strategically to influence the technology choice of the upstream firms. Richards (2003) finds that firms may have an incentive to pay an exorbitant price for the target to deter rival bidders when there is information asymmetry. Qiu and Zhou (2007) find that some firms merge strategically to trigger more mergers between rivals. Our paper highlights two strategic motives of mergers: (i) Mergers can be conducted strategically to preempt mergers at another tier; (ii) firms may open up a merger opportunity to deter mergers between suppliers or between customers.

Finally, our paper contributes to the literature on merger waves, particularly inter-industry merger waves. There are theoretical papers that focus on intra-industry merger waves (e.g., Fauli-Oller, 2000; Nilssen & Sørgard, 1998; Qiu & Zhou, 2007), and there are also empirical papers that study inter-industry merger waves (e.g., Ahern & Harford, 2014; Bhattacharyya & Amrita, 2011; Oberg & Holtstrom, 2006). However, there is not much theoretical work on inter-industry merger waves. Yao and Zhou (2015) study endogenous merger waves in vertically-related industries where firms may engage in both vertical and horizontal mergers. They show that whether and how firms merge depends crucially on the balance between vertical and horizontal externalities, and the

[5]The deterrent value of the merger option is also a notion of option value, that is, a firm's willingness to pay for the option. The preserving cost can be interpreted as the payment for the option.

[6]It is trivial that when both methods work, firms will carry out the merger if and only if the merger is profitable.

balance between upstream and downstream competition. Qiu and Zhou (2007) find that the strategic interaction between mergers contributes to intra-industry merger waves. We find the opposite for inter-industry merger waves, that is, the strategic preventive motive makes merger waves along a supply chain less likely.

3 | MODEL SETUP: TWO-STAGE GAME

We consider a two-tier supply chain model, that is, a supply chain with an upstream tier indexed by u and a downstream tier indexed by d. In each tier $k \in \{u, d\}$, there are $n_k \geq 3$ firms with identical marginal cost c_k initially.[7] To focus on the inter-tier strategic interaction between mergers in different tiers along the supply chain and to identify the mechanism how firms in one tier can use their merger options to prevent a merger in another tier, we assume that there is only one potential two-firm merger in each tier to abstract away from the intra-tier interaction between mergers within the same tier for simplicity. In tier k, the merger generates a synergy in the form of a cost reduction, that is, the merged firm has a marginal cost $c_k - \delta_k$ for some $\delta_k > 0$. We assume that merger synergy (δ_u, δ_d) is public information before the merger is conducted.

The game proceeds in two stages.

3.1 | Merger stage: Sequential game

In the first stage, also referred to as the merger stage, firms that have merger options decide whether to merge or not. We model the merger stage as a two-player sequential game since mergers are observable and irreversible. Each merging pair (i.e., the two firms who jointly decide whether to merge or not) is referred to as a single player or mover. One out of the two players is drawn randomly to move first, and the two players make their merger decisions sequentially. The sequential game ends after an endogenous number of rounds. The game ends when the current mover has merged before or has not merged but faces a state that he has faced before (i.e., indicates a repetition). Therefore, the game ends in one of three cases:

(1) The first mover merges in the first round, and the second mover makes his choice (either merge or not merge) in the second round. Then since the first mover has merged before, the game ends;
(2) The first mover does not merge in the first round, and the second mover does not merge either in the second round. Then since the first mover faces a state that he has faced before, the game ends;

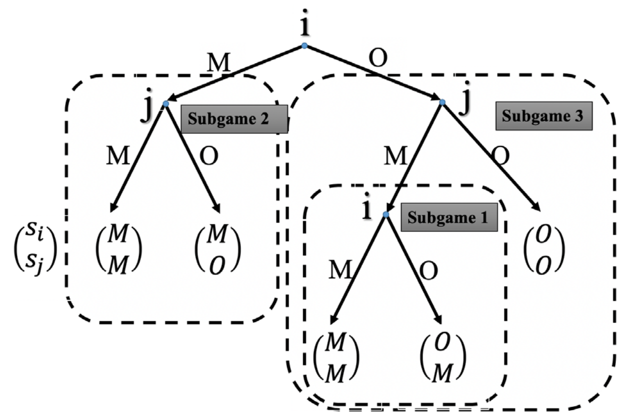


FIGURE 1 The merger game. We index the first mover by i and the second mover by j.

(3) The first mover does not merge in the first round, and the second mover merges in the second round. Then the first mover faces a different state and thus has one more chance to decide in the third round. Then since the second mover has merged before, the game ends.

The game tree is shown in Figure 1. In the two-player game, for player k, we index the other player by $-k$. In each round, the current mover k who has not merged decides $s_k \in \{M_k, O_k\}$ given the other player's state $s_{-k} \in \{M_{-k}, O_{-k}\}$, where M stands for decision "merge" or state "merged", and O for decision "not merge" or state "not merged". With slight abuse of notation, we use the same notation for both choices and states of players.

3.2 | Production stage: Successive Cournot competition

In the second stage, also referred to as the production stage, firms choose production quantities and receive profits given the market structure $\mathbf{s} = s_u s_d =\in \{O_u O_d, O_u M_d, M_u O_d, M_u M_d\}$. A homogeneous intermediate good (the input) is produced by firms in the upstream tier and is transformed into a homogeneous final good (the output) by firms in the downstream tier on a one-for-one basis. Firms in the two tiers interact through successive oligopoly.[8] Specifically, the downstream firms take the (endogenous) input price t as given, and choose their output quantities in a Cournot competition. The resulting total quantity, $Q(t)$, is then inverted to generate the inverse demand function for the input, $t(Q)$. We assume that the demand function for the final product is linear, $p(Q) = a - bQ$.

4 | ANALYSIS OF THE TWO-STAGE GAME

In this section, we solve the equilibrium of the two-stage game by backward induction.

[7]We impose the restriction $n_k \geq 3$ for the reason that if $n_k = 2$, a two-firm horizontal merger leads to a monopoly and thus is usually banned by anti-trust law.

[8]Successive oligopoly is a more appropriate setup for a homogeneous input, while bilateral oligopoly is more appropriate for heterogeneous, especially firm-specific, inputs. See Ghosh et al. (2014).

4.1 | Analysis of production stage

For a thorough understanding of successive Cournot competition, we consider a more general setting where a supply chain has an arbitrary number of tiers, each tier has an arbitrary number of firms, and firms in each tier may differ in marginal costs. Specifically, there are $M \geq 2$ tiers and $N_k \geq 1$ firms in tier $k \in \{1, \ldots, M\}$. Firm $f \in \{1, \ldots, N_k\}$ in tier k has a constant marginal cost $c_{k,f}$. We rank tiers from the downstream to the upstream, from 1 to M. Let t_k denote the price of output by firms in tier k, $Q = Q_{k-1}(t_k)$ denote the demand function for output by tier k from buyers in tier $k-1$, and $t_k = t_k(Q)$ denote the corresponding inverse demand function. Given the inverse demand function from consumers, $p(Q) = t_1(Q) = a_1 - b_1 Q$, we can derive inverse demand functions for other tiers *by mathematical induction* as follows.

For firm f in tier k, given inverse demand function $t_k(Q) = a_k - b_k Q$ and input price t_{k+1}, firm f chooses quantity $q_{k,f}$ to maximize its profit given belief about rival firms' quantities, as follows:

$$q_{k,f} = \arg\max_{q} \left\{ \left(t_k\left(q + \sum_{j \neq f} q_{k,j} \right) - t_{k+1} - c_{k,f} \right) q \right\}. \quad (1)$$

Then, from the first order condition (FOC), we can solve $q_{k,f}$ as a function of t_{k+1}. By summing up the order quantities of individual firms in tier k, we have the demand function for output produced by firms in tier $k+1$, that is,

$$Q = Q_k(t_{k+1}) = \sum_f q_{k,f}(t_{k+1}) = \frac{N_k}{N_k + 1} \frac{a_k - t_{k+1} - \bar{c}_k}{b_k}, \quad (2)$$

where $\bar{c}_k \equiv \frac{1}{N_k} \sum_f c_{k,f}$ is the average cost in tier k.

By inverting the demand function above, we get the inverse demand function as follows:

$$t_{k+1} = t_{k+1}(Q) = \left(a_k - \bar{c}_k\right) - \frac{N_k + 1}{N_k} b_k Q. \quad (3)$$

Then we have

$$\begin{cases} a_{k+1} = a_k - \bar{c}_k = a_1 - \sum_{j=1}^{k} \bar{c}_j, \\ b_{k+1} = \frac{N_k+1}{N_k} b_k = \left(\Pi_{j=1}^{k} \frac{N_j+1}{N_j} \right) b_1. \end{cases} \quad (4)$$

For tier M, there is no input, that is, $t_{M+1} = 0$, and we solve the equilibrium total quantity as follows:

$$Q^* = \frac{N_M}{N_M + 1} \frac{a_M - t_{M+1} - \bar{c}_M}{b_M} = \frac{\left(a_1 - \sum_{k=1}^{M} \bar{c}_k\right)}{\left(\Pi_{k=1}^{M} \frac{N_k+1}{N_k}\right) b_1}. \quad (5)$$

Given the equilibrium total quantity Q^*, we can solve equilibrium input prices $t_k^* = t_k(Q^*)$ and individual firms' profits $\pi_{k,f}$.[9]

Let's define notations as follows: $v_{k,f} \equiv (\bar{c}_k - c_{k,f})$ is firm f's cost advantage over its rivals, $A_k \equiv a_k - t_{k+1} - \bar{c}_k$ is tier k's markup potential, $A \equiv a - \sum_{k=1}^{M} \bar{c}_k$ is the supply chain's markup potential. Then, we have the general profit formula in the production stage.

Lemma 1 (Profit formula). *For a supply chain with $M \geq 2$ tiers and $N_k \geq 1$ firms in tier $k \in \{1, \ldots, M\}$, given inverse demand function from consumers $p(Q) = a - bQ$, the profit of firm f in tier k is*

$$\pi_{k,f} = \frac{1}{b_k} \left[\frac{1}{N_k + 1} A_k + v_{k,f} \right]^2, \quad (6)$$

$$\begin{cases} A_k = \left(\Pi_{j=k+1}^{M} \frac{N_j}{N_j+1} \right) A, \ k = 1, \ldots, M-1, \\ A_M = A, \\ b_k = \left(\Pi_{j=1}^{k-1} \frac{N_j+1}{N_j} \right) b, \quad k = 2, \ldots, M, \\ b_1 = b. \end{cases}$$

For our two-tier setting, $M = 2$, $k \in \{u, d\}$, let t denote the price of the output by the upstream tier. Then the inverse demand function and the equilibrium input price are as follows:

$$t(Q) = t_2(Q) = \left(a - \bar{c}_d\right) - \frac{N_d + 1}{N_d} bQ. \quad (7)$$

$$t^* = t(Q^*) = \frac{N_u}{N_u + 1} \bar{c}_u + \frac{1}{N_u + 1} (a - \bar{c}_d) \quad (8)$$

Given the general profit formula (6) in Lemma 1, we have

$$\begin{cases} \pi_{u,f} = \Pi_u(N_u, N_d, A, v_{u,f}) \\ \quad = \frac{N_d}{(N_d+1)b} \left[\frac{1}{N_u + 1} A + v_{u,f} \right]^2, \\ \pi_{d,f} = \Pi_d(N_u, N_d, A, v_{d,f}) \\ \quad = \frac{1}{b} \left[\frac{1}{N_d + 1} \frac{N_u}{N_u+1} A + v_{d,f} \right]^2. \end{cases} \quad (9)$$

With the profit formula $\Pi_u()$ and $\Pi_d()$ in (9), we can derive not only merging firms' profit, but also other firms' profit. Then we can analyze the effect of a merger on the merging firms and other firms. We impose the restriction that the merger synergy is not too high such that no firm exits due to the merger, otherwise the merged firm's cost advantage is so strong that its rivals in the same tier are unable to survive the competition.[10]

For the sequential merger game, let $T_k(\mathbf{s})$ denote player k's payoff if the equilibrium outcome is $\mathbf{s}$ after the merger stage. The payoff $T_k(\mathbf{s})$ is derived by applying the profit formula in (9) and is summarized in Table 1. For illustration, let us consider the subgame $s_u s_d = M_u O_d$, that is, only the upstream merger is carried out in the merger stage. Due to the upstream merger, the number of firms in the upstream tier decreases by 1, $N_u(M_u) = n_u - 1$, the chain's markup potential increases,

[9] For derivation details, see online appendix.

[10] A necessary but insufficient condition for no exit caused by a merger is $\delta_k < A(O_u O_d) = a - c_u - c_d$. See online appendix for proof.

TABLE 1 Payoff of the two-player merger game.

$s_u s_d$	T_u	T_d
$O_u O_d$	$\frac{2}{b}\frac{n_d}{n_d+1}\left[\frac{1}{n_u+1}(a-c_u-c_d)\right]^2$	$\frac{2}{b}\left[\frac{n_u}{n_u+1}\frac{1}{n_d+1}(a-c_u-c_d)\right]^2$
$M_u O_d$	$\frac{1}{b}\frac{n_d}{n_d+1}\left[\frac{1}{n_u}\left(a-c_u-c_d+\frac{\delta_u}{n_u-1}\right)+\frac{n_u-2}{n_u-1}\delta_u\right]^2$	$\frac{2}{b}\left[\frac{n_u-1}{n_u}\frac{1}{n_d+1}\left(a-c_u-c_d+\frac{\delta_u}{n_u-1}\right)\right]^2$
$O_u M_d$	$\frac{2}{b}\frac{n_d-1}{n_d}\left[\frac{1}{n_u+1}\left(a-c_u-c_d+\frac{\delta_d}{n_d-1}\right)\right]^2$	$\frac{1}{b}\left[\frac{n_u}{n_u+1}\frac{1}{n_d}\left(a-c_u-c_d+\frac{\delta_d}{n_d-1}\right)+\frac{n_d-2}{n_d-1}\delta_d\right]^2$
$M_u M_d$	$\frac{1}{b}\frac{n_d-1}{n_d}\left[\frac{1}{n_u}\left(a-c_u-c_d+\frac{\delta_d}{n_d-1}+\frac{\delta_u}{n_u-1}\right)+\frac{n_u-2}{n_u-1}\delta_u\right]^2$	$\frac{1}{b}\left[\frac{n_u-1}{n_u}\frac{1}{n_d}\left(a-c_u-c_d+\frac{\delta_d}{n_d-1}+\frac{\delta_u}{n_u-1}\right)+\frac{n_d-2}{n_d-1}\delta_d\right]^2$

$A(M_u, O_d) = A(O_u, O_d) + \delta_u/(n_u - 1)$, and the merged firm gains a cost advantage over its rivals, $v_u(M_u) = \overline{c}_u - (c_u - \delta_u) = \frac{n_u-2}{n_u-1}\delta_u$. The payoff of the upstream player is the profit of the merged firm, while the payoff of the downstream player is the sum of the profits of the two firms that have not merged, that is,

$$\begin{cases} T_u(M_u O_d) = \Pi_u\left(n_u - 1, n_d, a - c_u - c_d + \frac{\delta_u}{n_u-1}, \frac{n_u-2}{n_u-1}\delta_u\right), \\ T_d(M_u O_d) = 2 \times \Pi_d\left(n_u - 1, n_d, a - c_u - c_d + \frac{\delta_u}{n_u-1}, 0\right). \end{cases} \tag{10}$$

In sum, $N_k(s_k)$, player k's cost advantage $v_k(s_k)$, supply chain's markup potential $A(s_u, s_d)$ depend on s_k in the following way:

$$\begin{cases} N_k(M_k) &= n_k - 1, \\ N_k(O_k) &= n_k, \\ A(M_k, s_{-k}) &= A(O_k, s_{-k}) + \frac{\delta_k}{n_k-1}, \\ A(O_u O_d) &= a - c_u - c_d, \\ v_k(M_k) &= \frac{n_k-2}{n_k-1}\delta_k, \\ v_k(O_k) &= 0. \end{cases} \tag{11}$$

By substituting (11) into profit formula (9), we solve the payoff for all subgames. Then, the payoff for the two players in the merger game is shown in Table 1.

4.2 | Analysis of merger stage

Given the payoff in Table 1, we can view the merger stage as a complete sequential game. We first identify a motive for firms to prevent a merger among suppliers or customers and then solve the merger game.

4.2.1 | Identification of preventive motive

To identify the strategic motive, we address the following three questions about a horizontal merger in a supply chain: (i) Under what condition is a merger profitable? (ii) Under what condition is a merger beneficial to firms at another tier? (iii) Under what condition do firms at one tier want to precipitate or prevent a merger at another tier? The answer to the third question is a natural result of the answers to the first two questions. Firms at one tier want to prevent a merger at another tier if and only if the merger is profitable but harmful to them, while firms want to precipitate a merger if and only if the merger is unprofitable but beneficial to them.

In the two-player merger game, holding the other player's choice s_{-k}, we say that merger k is *profitable* if the merger increases player k's payoff, that is, the *own-effect* of merger k, $T_k(M_k s_{-k}) - T_k(O_k s_{-k})$, is positive, and we say that merger k is *beneficial* to the other player if it increases the other player's payoff, that is, the *cross-effect* of merger k, $T_{-k}(M_k s_{-k}) - T_{-k}(O_k s_{-k})$, is positive. We can derive the condition for merger k to be profitable or beneficial to the other player as follows:

$$\underbrace{T_k(M_k, s_{-k}) - T_k(O_k, s_{-k})}_{\text{own-effect of merger } k} > 0 \Leftrightarrow \delta_k > \delta_k^W(s_{-k}),$$
$$s_{-k} \in \{O_{-k}, M_{-k}\}. \tag{12}$$

$$\underbrace{T_{-k}(M_k, s_{-k}) - T_{-k}(O_k, s_{-k})}_{\text{cross-effect of merger} k} > 0 \Leftrightarrow \delta_k > \delta_k^R(s_{-k}),$$
$$s_{-k} \in \{O_{-k}, M_{-k}\}. \tag{13}$$

Lemma 2. *In a two-tier supply chain, given the other player's choice s_{-k}, merger k is profitable if $\delta_k > \delta_k^W(s_{-k})$, and is beneficial to the other player if $\delta_k > \delta_k^R(s_{-k})$, where*

$$\begin{cases} \delta_u^W(s_d) = \frac{\left(\sqrt{2}-1\right)n_u-1}{n_u^2-1}A(O_u, s_d), \\ \delta_d^W(s_u) = \frac{(n_d-1)^2}{((n_d-1)^2-1)\frac{N_u(s_u)+1}{N_u(s_u)}+1}\frac{\left(\sqrt{2}-1\right)n_d-1}{n_d^2-1}A(s_u, O_d), \\ \delta_u^R(s_d) = \frac{1}{n_u+1}A(O_u, s_d), \\ \delta_d^R(s_u) = \frac{1}{n_d+1}\frac{\sqrt{n_d^2-1}}{n_d+\sqrt{n_d^2-1}}(A(s_u, O_d) + N_u(s_u)v_u(s_u)). \end{cases} \tag{14}$$

Throughout this paper, there are many synergy thresholds. First, we use subscript k to denote the threshold of player k's merger synergy, and use superscript to denote different types of thresholds. For example, δ_k^W is the threshold of synergy δ_k above which merger k is profitable, and δ_k^R is the threshold of δ_k above which merger k is beneficial to the other player. Second, to avoid too many notations, we use the same notation for function names and variable names. For example, δ_k^W depends on the other player's merger decision $s_{-k} \in (O_{-k}, M_{-k})$. When it is necessary to explicate such dependence, we write $\delta_k^W(s_{-k})$. Table 2 shows definitions and characterizations of all synergy thresholds.

TABLE 2 Thresholds of merger synergies.

Threshold	Definition and characterization
$\delta_k^W(s_{-k})$	Threshold of δ_k above which merger k is profitable given player $-k$'s choice s_{-k}. $T_k(M_k,s_{-k})-T_k(O_k,s_{-k})>0 \Leftrightarrow \delta_k>\delta_k^W(s_{-k})$.
$\delta_k^R(s_{-k})$	Threshold of δ_k above which merger k benefits player $-k$ given player $-k$'s choice s_{-k}. $T_{-k}(M_k,s_{-k})-T_{-k}(O_k,s_{-k})>0 \Leftrightarrow \delta_k>\delta_k^R(s_{-k})$.
$\delta_k^C(\delta_{-k})$	Threshold of δ_k above which player k will merge if merger k is a complement to merger $-k$. $T_k(M_kM_{-k})-T_k(O_kO_{-k})>0 \Leftrightarrow \delta_k>\delta_k^C(\delta_{-k})$.
$\delta_k^S(\delta_{-k})$	Threshold of δ_k above which player k will merge if merger k is a substitute for the other merger. $T_k(M_kO_{-k})-T_k(O_kM_{-k})>0 \Leftrightarrow \delta_k>\delta_k^S(\delta_{-k})$.
δ_k^*	Threshold of δ_k above which merger k tends to discourage merger $-k$. $\delta_{-k}^W(M_k)>\delta_{-k}^W(O_k) \Rightarrow \delta_k>\delta_k^*$.
$\delta_k^D(\delta_{-k})$	Threshold of δ_k above which carrying out merger k can discourage player $-k$ from merging. $\delta_{-k}^W(M_k)>\delta_{-k} \Rightarrow \delta_k>\delta_k^D(\delta_{-k})$.
$\delta_k^{(1)}(\delta_{-k})$	Threshold of δ_k above which merger k is a credible threat. $\delta_k>\delta_k^W(M_{-k}) \Rightarrow \delta_k>\delta_k^{(1)}(\delta_{-k})$.
$\delta_k^{(2)}(\delta_{-k})$	Threshold of δ_k below which merger k inflicts a harm on player $-k$ that outweighs player $-k$'s gain from its own merger. $\delta_{-k}<\delta_{-k}^C(\delta_k) \Rightarrow \delta_k<\delta_k^{(2)}(\delta_{-k})$.
δ_{-k}^H	Threshold of δ_{-k} below which it is possible for merger k to deter merger $-k$. $\delta_k^{(1)}(\delta_{-k})<\delta_k^{(2)}(\delta_{-k}) \Rightarrow \delta_{-k}<\delta_{-k}^H$.
δ_{-k}^L	Threshold of δ_{-k} above which player k will merge passively as the deterrence effect disappears. $\delta_k^{(2)}(\delta_{-k})<\delta_k^C(\delta_{-k}) \Rightarrow \delta_{-k}>\delta_{-k}^L$.

We can decompose the effect of a merger into the effect of a reduction in the number of firms (*concentration effect*) and the effect of a cost reduction (*efficiency effect*).

First, consistent with the previous literature, we find that a horizontal merger in a supply chain is profitable if and only if the synergy is high enough. A merger without synergy will reduce competition and therefore contribute to the expansion of its rivals, which in turn hurts the merging firms.[11] This free-riding effect is first identified in Salant et al. (1983). It is intuitive that the merger synergy increases the merged entity's profit, which benefits the merging firms. Thus, merger k is profitable if and only if the synergy is higher than some threshold denoted by δ_k^W, which we call *profitable threshold*.

Second, a merger benefits firms in another tier if and only if the synergy is high enough. Under successive oligopoly, firms in the upstream tier affect firms in the downstream tier through the equilibrium input price t while firms in the downstream tier affect firms in the upstream tier through the inverse demand curve $t(Q)$. From the profit formula in (6), we find that a more efficient tier (a tier with a smaller average marginal cost) benefits firms in another tier, while a more concentrated tier (a tier with fewer firms) hurts them. Downstream consolidation will reduce the demand for the input for any given input price, and thus is a negative demand shock to firms in the upstream tier. Upstream consolidation will reduce the supply and increase the equilibrium input price, and thus is a negative supply shock to firms in the downstream tier. Therefore, a merger without synergy is always harmful to firms in another tier. A cost reduction in the upstream industry decreases the equilibrium input price, which benefits the downstream firms. A more efficient firm in the downstream industry has a higher willingness to pay for the input, which increases the downstream industry's willingness to pay for input, $t(Q)$, and thus benefits the upstream firms. Thus, merger k is beneficial to firms in another tier if and only if the synergy is higher than some threshold denoted by δ_k^R, which we call *beneficial threshold*.

In sum, when the merger synergy is high enough, the efficiency effect dominates the concentration effect, and the merger will benefit the merging firms as well as all firms in its up- or downstream tier. A natural question is which of the two thresholds is higher, and what the implication is.

Lemma 3. *It is more demanding for a merger to be beneficial to firms in another tier than to be profitable itself, that is, the beneficial threshold is higher than the profitable threshold,*

$$\delta_k^R(s_{-k})>\delta_k^W(s_{-k}),\quad s_{-k}\in\{O_{-k},M_{-k}\},k\in\{u,d\}. \tag{15}$$

Lemma 3 states that it requires a higher synergy for a merger to benefit firms in another tier than to benefit itself. We can prove that for any demand curve, $\delta_u^R > \delta_u^W$; or equivalently, an upstream merger with synergy $\delta_u = \delta_u^R$ is profitable. If $\delta_u = \delta_u^R$, the downstream firms are unaffected after the upstream merger, which implies that the input price remains unchanged. According to Cournot competition, it further implies that the production quantity by the merged entity is the same as the total production quantities by the two merging firms before the merger. Due to a cost reduction, the upstream merged firm is strictly better off, that is, the merger is profitable. However, for the comparison of the two synergy thresholds of a downstream merger, the argument is complicated. The intuition is that merger synergy gives the merged firm a direct competitive advantage over competing firms, while it benefits the other industry indirectly through the inverse demand curve.

Proposition 1 (Preventive motive). *It is possible for a firm in one tier to have an incentive to prevent a merger in another tier, but impossible for the firm to have an incentive to precipitate the merger.*

[11]For a general demand function, a downstream merger without synergy always reduces aggregate output regardless of the effect on input prices. Under linear demand, such a merger does not affect the input price as shown in Ghosh et al. (2014), and thus a downstream merger without synergy is unprofitable due to the free-riding effect.

TABLE 3 Equilibrium outcome in the merger game.

Condition		Outcome
$\delta_u > \delta_u^W(M_d)$ and $\delta_d > \delta_d^W(M_u)$ and $(\delta_d > \delta_d^C$ or $\delta_u > \delta_u^C)$		M_uM_d
$\delta_u > \delta_u^W(O_d)$ and $\delta_d < \delta_d^W(M_u)$	$\delta_u > \delta_u^W(M_d)$ or $\delta_d < \delta_d^S$	M_uO_d
	$\delta_u < \delta_u^W(M_d)$ and $\delta_d > \delta_d^S$	Only the first mover merges
$\delta_u < \delta_u^W(M_d)$ and $\delta_d > \delta_d^W(O_u)$		O_uM_d
Otherwise		O_uO_d

Combining Lemmas 2 and 3 together, we can infer that for a specific firm in a supply chain, given a merger in another tier: (i) if the merger is profitable, it may be harmful to the firm (when the synergy falls between the two thresholds) and thus the firm has an incentive to do something strategically to prevent it; (ii) if the merger is unprofitable, the merger must be harmful to the firm and thus the firm will have no incentive to precipitate it.

As we will see in the following analysis, the preventive motive identified in Proposition 1 plays a pivotal role in the players' interaction of the merger stage game.

4.2.2 | Equilibrium outcome of merger stage

Hereafter, we focus on the merger stage. We treat the merger stage as a complete sequential game with the payoff table $T_k(\mathbf{s})$ in Table 1 and the game tree shown in Figure 1.

In the sequential game, a complete strategy profile should specify the decision of each player when the player has not merged and it is his turn to decide whether to merge or not given the other player's state. Let $R_k(s_{-k}) \in \{M_k, O_k\}$ denote player k's equilibrium decision of whether or not to merge given the other player's state $s_{-k} \in \{M_{-k}, O_{-k}\}$. When it is player k's turn to decide, we can classify the decision-making problem into three cases according to the strategic effect of player k's current choice on the other player's choice. In each case, player k will merge if its merger synergy is high enough, that is, higher than some threshold.

Case (a) No strategic effect. In this case, player k's decision in this round has no effect on player $-k$'s merger decision in the end. In the two-player game, note that $\delta_k^W(s_{-k})$ denote player k's profitable threshold given the other player's state s_{-k}. If player $-k$ has not merged and will never merge later whatever player k's choice is, then player k will merge if and only if $\delta_k > \delta_k^W(O_{-k})$; If player $-k$ has merged, or player $-k$ has not merged but will merge later whatever player k's current choice is, then player k will merge if and only if $\delta_k > \delta_k^W(M_{-k})$. The thresholds are characterized by (12).

Case (b) Strategic complement. If player $-k$ has not merged yet and will merge if and only if player k merges, we say that player k's merger is a strategic complement to player $-k$'s merger. We denote the synergy threshold by δ_k^C, which is characterized as follows:

$$
\begin{aligned}
T_k(M_kM_{-k}) - T_k(O_kO_{-k}) &= \underbrace{T_k(M_kO_{-k}) - T_k(O_kO_{-k})}_{\text{own-effect of merger } k} \\
&\quad + \underbrace{T_k(M_kM_{-k}) - T_k(M_kO_{-k})}_{\text{cross-effect of merger } -k} > 0 \\
&\Leftrightarrow \delta_k > \delta_k^C.
\end{aligned} \tag{16}
$$

Case (c) Strategic substitute. If player $-k$ has not merged yet and will merge if and only if player k does not merge, we say that player k's merger is a strategic substitute for player $-k$'s merger. We denote the synergy threshold by δ_k^S, which is characterized as follows:

$$
\begin{aligned}
T_k(M_kO_{-k}) - T_k(O_kM_{-k}) &= \underbrace{(T_k(M_kO_{-k}) - T_k(O_kO_{-k}))}_{\text{own-effect of merger } k} \\
&\quad - \underbrace{(T_k(O_kM_{-k}) - T_k(O_kO_{-k}))}_{\text{cross-effect of merger } -k} > 0 \\
&\Leftrightarrow \delta_k > \delta_k^S.
\end{aligned} \tag{17}
$$

By comparing the thresholds in the three cases, we can figure out the equilibrium outcome and the corresponding condition, which are summarized in the Table 3.

According to the three cases discussed above, players make decisions based on the signs of Equations (12), (16), and (17), that is, player k compares δ_k with thresholds $(\delta_k^W, \delta_k^C, \delta_k^S)$. Figure 2 shows the equilibrium outcome in the (δ_u, δ_d) parameter space. At the four corners of the diagram, that is, each player's merger synergy is either very high or very low and thus the two players' decisions have no strategic effect on each other, $M_uO_d/M_uM_d/O_uO_d/O_uM_d$ is the equilibrium outcome. When merger synergies are moderate, that is, in the central part of the diagram, there may be strategic interactions, which is the focus of our discussion. When one player's merger has a strategic effect on the other player's decision in the merger game, a strategic-thinking manager has to take into consideration the feedback effect of the other player's decision to merge or not following his own merger, besides the immediate effect of his own merger.

5 | MODEL IMPLICATIONS

In this section, we discuss the model implications for how to prevent a merger between suppliers or customers. There are many interesting managerial insights. For example, we point out that it is possible for an acquiring firm to choose a target firm with a smaller synergy for deterrence purpose. In the end, we also discuss from a policymaker's perspective about

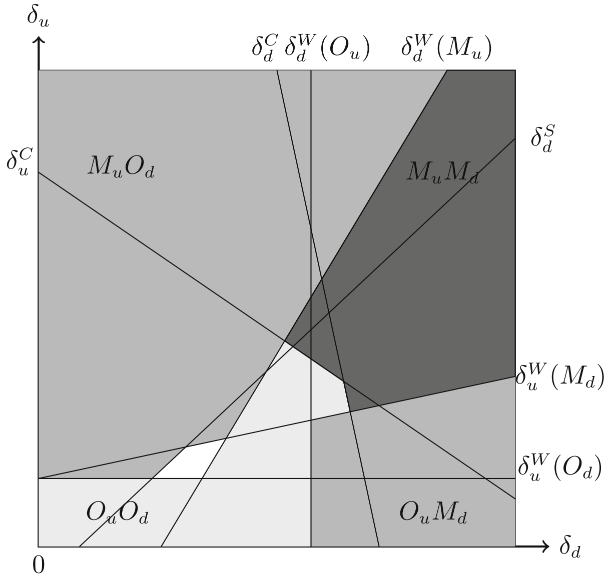


FIGURE 2 Equilibrium outcome in the merger game. In the white area, the first mover merge preemptively and the second mover does not merge; in the rest of the parameter space, the equilibrium outcome does not depend on who moves first.

whether the preventive methods conflict with consumers' welfare.

5.1 | Implications: Two preventive methods

In the equilibrium outcome, we find two interesting cases where one player withholds a profitable merger or conducts an unprofitable merger strategically to prevent the other player from merging. We thoroughly examine each case in the following discussion and show how the preventive motive plays a role.

For player k, when it faces a profitable merger by the other player, that is, $\delta_{-k} > \delta_{-k}^W(O_k)$, player k can prevent the other player from merging in two ways – either discourage the other player from merging by carrying out its merger or deter the other player from merging by preserving its merger option. The first method requires $R_{-k}(M_k) = O_{-k}$ while the second method requires $R_{-k}(O_k) = O_{-k}$.

5.1.1 | Method one: One player discourages another merger by merging

We say that one shock weakens the merger incentive, or equivalently speaking, tends to *discourage* mergers, if the shock increases the profitable threshold. If a profitable merger becomes unprofitable after the shock, that is, the merger synergy falls between the two profitable thresholds before and after the shock, we say that the shock effectively discourages the merger. Similarly, we say that a shock tends to *encourage* mergers if the shock decreases the profitable threshold. Given a fixed level of merger synergy, if a shock tends to increase the markup (i.e., price minus unit cost) of all firms in a tier, then the ratio of the merged firm's markup to its rivals' markup becomes smaller, that is, the synergy gives the merged firm a smaller competitive advantage over its rivals. Thus, it requires a higher synergy to overcome the free-riding effect, that is, the profitable threshold increases. Therefore, a shock that increases the overall markup in a tier tends to discourage mergers in that tier.

Lemma 4. *Merger k tends to discourage merger $-k$, that is, $\delta_{-k}^W(M_k) > \delta_{-k}^W(O_k)$, if $\delta_k > \delta_k^*$ for some threshold δ_k^*, where*

$$\begin{cases} \delta_u^* = \frac{(n_d-2)n_d}{(n_d-2)n_d(n_u+1)+n_u} A(O_uO_d); \\ \delta_d^* = 0. \end{cases} \tag{18}$$

Back to the two-merger setup in a two-tier supply chain, a merger in tier $k \in \{u, d\}$ can be interpreted as a shock to the other tier, which affects the profitable threshold of mergers in the other tier through two channels, the concentration effect and the efficiency effect. First, the concentration effect of a merger harms firms in another tier, which tends to decrease the markup in that tier and thus tends to encourage mergers in that tier. Second, the efficiency effect of a merger benefits firms in another tier and thus tends to discourage mergers. Therefore, a merger tends to discourage mergers in another tier as long as the efficiency effect dominates, that is, its synergy is high enough.[12] Additionally, the discouraging effect is stronger if the synergy is higher. That is, the profitable threshold in another tier after merger k is conducted increases in the synergy of merger k, that is, $\delta_{-k}^W(M_k)$ increases in δ_k. It implies that, given any fixed level of $\delta_{-k} > \delta_{-k}^W(O_k)$, player k can always merge preemptively to discourage the other merger as long as δ_k is high enough such that $\delta_{-k}^W(M_k) > \delta_{-k}$. In Figure 3, the open triangular area denoted by S_k represents the parameter space $\delta_{-k}^W(O_k) < \delta_{-k} < \delta_{-k}^W(M_k)$, that is, the parameter space within which player k can discourage the other player from merging by carrying out his own merger.

Proposition 2 (Discouraging method). *Given merger $-k$ with synergy $\delta_{-k} > \delta_{-k}^W(O_k)$, merger k with synergy δ_k, if conducted, can discourage merger $-k$ as long as δ_k is high enough relative to δ_{-k} such that $\delta_{-k}^W(M_k) > \delta_{-k}$, that is, $\delta_k > \delta_k^D(\delta_{-k})$.*

5.1.2 | Method two: One player deters another merger by reserving its merger option

An important implication of Lemma 3 is that a profitable merger can inflict significant damage on firms in another

[12] For any demand curve, the concentration effect of the upstream merger will always increase the input price. Thus, it requires a positive synergy for the efficiency effect to offset the concentration effect, that is, $\delta_u^* > 0$. However, the concentration effect of the downstream merger on the input price is less clear. More generally, Ghosh et al. (2014) prove that for demand curves with constant elasticity of slope including the linear demand, a downstream merger without synergy does not affect the input price. Thus, $\delta_d^* = 0$.

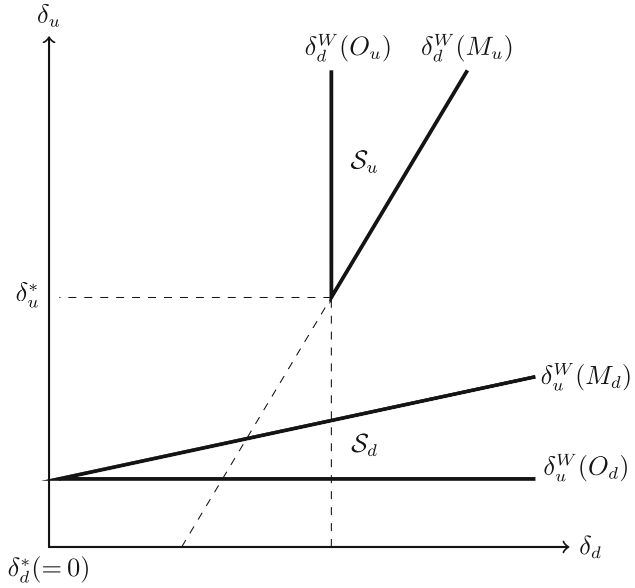


FIGURE 3 One merger discourages another merger.

tier. This suggests that a firm can use its merger option as a credible threat to their customers or suppliers to deter them from doing something harmful to the firm. Back to the two-merger game, for the merger option by player k to deter the other merger, that is, $R_{-k}(O_k) = O_{-k}$, it must meet two conditions simultaneously: credibility and severe damage to the other player. The first condition, credibility, requires the synergy of merger k to be high enough such that merger k is profitable provided merger $-k$ is conducted, that is,

$$\delta_k > \delta_k^W(M_{-k}) \Rightarrow \delta_k > \delta_k^{(1)}(\delta_{-k}). \tag{19}$$

The second condition requires both δ_k and δ_{-k} to be low enough such that the harm by merger k to the other player outweighs the other player's gain from merger $-k$; thus player $-k$ will not merge taking into consideration the feedback effect of merger k,

$$\delta_{-k} < \delta_{-k}^C(\delta_k) \Rightarrow \delta_k < \delta_k^{(2)}(\delta_{-k}). \tag{20}$$

Therefore, for merger option k to qualify as a deterrent, δ_k should be neither too high nor too low. Additionally, the two conditions (19) and (20) together implicitly require $\delta_k^{(1)}(\delta_{-k}) < \delta_k^{(2)}(\delta_{-k})$ to hold. Due to the discouraging effect of one merger on the other that we have analyzed before, $\delta_k^W(M_{-k})$, also referred to as $\delta_k^{(1)}(\delta_{-k})$, increases in δ_{-k}; given a higher δ_{-k}, it needs a lower δ_k for the harm that merger k inflicts on the other player to outweigh the other player's gain from his own merger, that is, $\delta_k^{(2)}(\delta_{-k})$ decreases in δ_{-k}. Thus, for $\delta_k^{(1)}(\delta_{-k}) < \delta_k^{(2)}(\delta_{-k})$ to hold, δ_{-k} must be low enough, that is,

$$\delta_k^{(1)}(\delta_{-k}) < \delta_k^{(2)}(\delta_{-k}) \Rightarrow \delta_{-k} < \delta_{-k}^H. \tag{21}$$

As δ_{-k} increases within $\left(\delta_{-k}^W(O_k), \delta_{-k}^H\right)$, the range $\left(\delta_k^{(1)}(\delta_{-k}), \delta_k^{(2)}(\delta_{-k})\right)$ shrinks, that is, it becomes more difficult for merger option k to deter the other merger.

Proposition 3 (Deterrence method). *Given merger $-k$ with synergy $\delta_{-k} > \delta_{-k}^W(O_k)$, only if δ_{-k} is not too high, it is possible for player k to deter player $-k$ from merging by preserving its merger option, that is, not carrying out its merger. Specifically,*

(i) there exists some threshold $\delta_{-k}^H > \delta_{-k}^W(O_k)$ such that $\delta_k^{(1)}(\delta_{-k}) < \delta_k^{(2)}(\delta_{-k})$ if $\delta_{-k} < \delta_{-k}^H$;

(i) if $\delta_k \in \left(\delta_k^{(1)}(\delta_{-k}), \delta_k^{(2)}(\delta_{-k})\right)$, then $R_{-k}(O_k) = O_{-k}$.

The Proposition 3 can be illustrated in Figure 4. If synergies are in $\mathcal{D}_k$, the decision of player $-k$ given O_k in the sequential game is $R_{-k}(O_k) = O_{-k}$. We call $\mathcal{D}_k$ the *deterrent area* for player k. If player k reserves its merger option, that is, does not merge but keep its merger option, then player $-k$ will withhold a seemingly profitable merger. In this case, both behave in a way to prevent the other merger: Player k deters the other merger by preserving its merger option, and the other player, facing the deterrent, will withhold the profitable merger to avoid triggering the harmful merger.

5.1.3 | Implications of two methods for two phenomena

Propositions 2 and 3 summarize the two potential methods that one player can use to prevent another merger, and for each method we indicate the corresponding parameter space within which this method works in Figures 3 and 4. The parameter space $\mathcal{S}_u \cup \mathcal{D}_u$ is above $\delta_u^W(M_d)$, which implies that a firm with an unprofitable merger option cannot prevent a merger between its customers; The parameter space $\mathcal{S}_d \cup \mathcal{D}_d$ crosses the cutoff line $\delta_d^W(M_u)$, which implies that a firm with an unprofitable merger option may prevent a merger between its suppliers.

When either method works for player k to prevent the other merger, that is, $(\delta_u, \delta_d) \in \mathcal{D}_k \cup \mathcal{S}_k$, would player k prevent the other merger, and if so, which method does he choose? We answer this question case by case as follows.

Case one: Both methods work. When both methods work, that is,$(\delta_u, \delta_d) \in \mathcal{D}_k \cap \mathcal{S}_k$, then $R_{-k}(s_k) = O_{-k}$ for $s_k \in \{O_k, M_k\}$. That is, player k's choice has no strategic effect on the other player's choice. Anticipating that the other player will not merge regardless of its merger decision, player k will merge if and only if merger k is profitable as in (12). For the downstream player, in $\mathcal{D}_d \cap \mathcal{S}_d$ in Figure 4, we see that the downstream player will reserve its merger option in area ③ and will conduct the merger otherwise. For the upstream player, we know that the upstream merger is profitable in $\mathcal{D}_u \cup \mathcal{S}_u$ and thus the upstream player will always merge in $\mathcal{D}_u \cap \mathcal{S}_u$.

Case two: Only the deterrence method works. When only the deterrence method works, that is, $(\delta_u, \delta_d) \in \mathcal{D}_k \setminus \mathcal{S}_k$, then $R_{-k}(O_k) = O_{-k}$ and $R_{-k}(M_k) = M_{-k}$. That is, player k's merger is a strategic complement to the other merger. Player k, anticipating that its merger is a strategic complement to the other

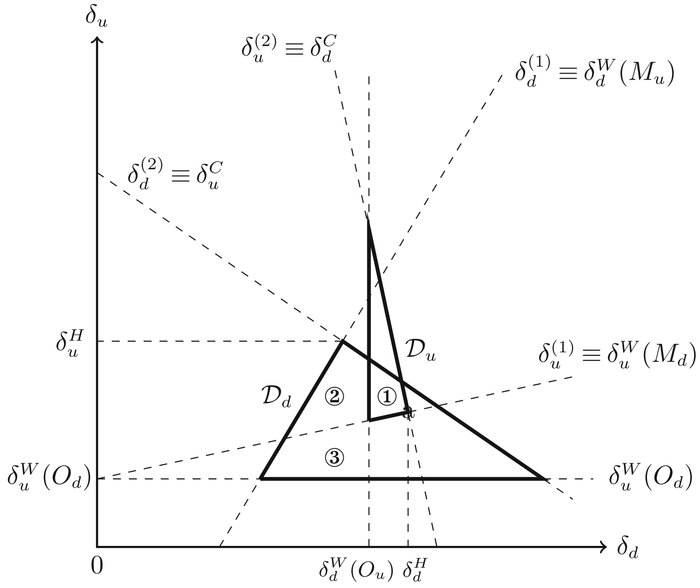


FIGURE 4 Effective deterrent area $(\mathcal{D}_u, \mathcal{D}_d)$.

merger, will merge if and only if $\delta_k > \delta_k^C$ as in Equation (16). For the downstream player, in $\mathcal{D}_d \setminus \mathcal{S}_d$ in Figure 4, it reserves its merger option in area ①-②; for the upstream player, in $\mathcal{D}_u \setminus \mathcal{S}_u$, it reserves its merger option in area ①. Note that area ① is the overlapping area of $\mathcal{D}_u$ and $\mathcal{D}_d$ where each player is subject to the other player's deterrent and reserves his own merger option. In the sequential game, the two players escape from the prisoners' dilemma, that is, a merger war is avoided through mutual deterrence.

> **Proposition 4** (Profitable merger withheld). *A profitable merger may be withheld due to the deterrence by the other player. Specifically, as shown in Figure 4,*
>
> *(i) in ①, each player withholds a profitable merger subject to the other player's merger threat;*
> *(ii) in ②-③, the upstream player, deterred by the downstream merger option, withholds a profitable merger.*

Case three: Only the discouraging method works. When only the discouraging method works, that is, $(\delta_u, \delta_d) \in \mathcal{S}_k \setminus \mathcal{D}_k$, then $R_{-k}(O_k) = M_{-k}$ and $R_{-k}(M_k) = O_{-k}$. That is, player k's merger is a strategic substitute for the other merger. Then player k decides between his own merger and the other merger and will choose to merge if and only if $\delta_k > \delta_k^S$ as in (17). For the upstream player, we can show that $\delta_u > \delta_u^S$ always holds in $\mathcal{S}_u \setminus \mathcal{D}_u$ and $\delta_u > \delta_u^W(O_d)$ always holds in $\mathcal{S}_u$. Thus, the upstream player will conduct the profitable merger. For the downstream player, there are two unconnected parts of $\mathcal{S}_d \setminus \mathcal{D}_d$, as shown in Figure 5. In the right part, the downstream merger is profitable and $\delta_d > \delta_d^S$ always holds. Thus, the downstream player will conduct the profitable merger. In the left part, the downstream merger is unprofitable and the upstream merger is harmful to the downstream player. Thus, the downstream player faces a choice between two undesirable mergers. In ④, if the downstream player moves first, he will choose to conduct his own merger, which is less undesirable than the merger by the upstream player. In ④, compared to ③, the downstream player has to incur the loss of an unprofitable merger to discourage the upstream merger since the merger threat is no longer credible. If the downstream player moves first, his choice switches from M to O as δ_d increases, from ④ to ③. This phenomenon is counterintuitive, where a player will merge when the synergy is low and not merge when the synergy is higher. The reason is that when the synergy is high enough, the merger option becomes credible and thus can deter the upstream merger. In ④, we see a first-mover advantage and each player's merger is a strategic substitute for the other merger. That is, the first mover merges, and the second mover does not merge and is harmed by the first mover's merger.

> **Proposition 5** (Unprofitable merger conducted). *An unprofitable merger may be conducted to discourage the other merger. Specifically, in Figure 5, in area ④, the downstream player, if he moves first, will conduct the unprofitable merger preemptively in order to discourage the upstream merger.*

From Propositions 4 and 5, we can conclude that it is the preventive motive that explains the phenomena where a profitable merger is withheld or an unprofitable merger is conducted.

5.2 | Implications for a choice between merger options

Thus far, our analysis is based on the assumption that the synergy of a potential merger is exogenous and publicly known even before the merger is completed. If a firm has multiple acquisition targets with different synergy levels,

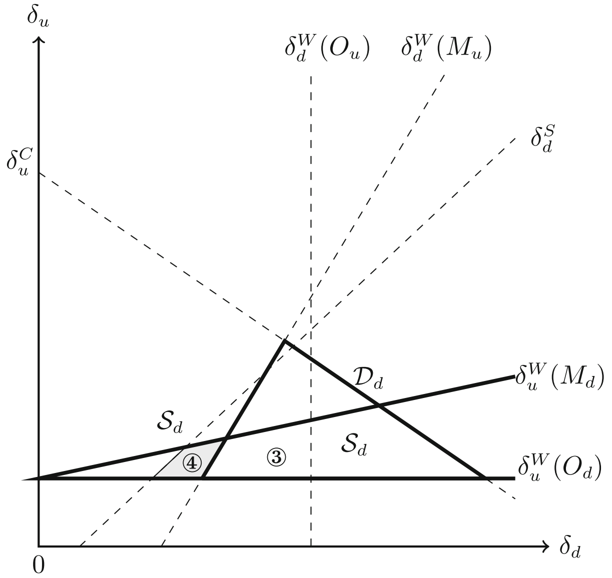


FIGURE 5 Two ways for a downstream player to prevent the profitable upstream merger. (i) in $\mathcal{D}_d$, the downstream player can deter the upstream merger by not merging; (ii) in $\mathcal{S}_d$, the downstream player can discourage the upstream merger by merging.

which target should the firm choose? If there is information asymmetry, that is, only merging partners (insiders) know their own synergy from the pre-merger talks while outsiders know the synergy only after the merger is completed, do firms have any incentive to cheat?

To answer these questions, we first study how one's payoff changes with one's merger synergy in the two-player merger game. We define the value of a merger option as follows:

$$V_k(\delta_k;\delta_{-k}) \equiv \tilde{T}_k(\delta_k;\delta_{-k}) - \tilde{T}_k(0;\delta_{-k}), \tag{22}$$

where $\tilde{T}_k(\delta_k;\delta_{-k}) \equiv T(s_k(\delta_k,\delta_{-k}), s_{-k}(\delta_k,\delta_{-k}))$ is player k's equilibrium payoff and $s_k(\delta_k,\delta_{-k})$ is player k's equilibrium outcome in the two-player merger game as functions of synergies (δ_k,δ_{-k}) holding other parameters fixed. Note that $\tilde{T}_k(0;\delta_{-k})$ equals the payoff of player k without the merger option. Thus, $V_k(\delta_k;\delta_{-k})$ is a notion of option value, that is, the total willingness to pay of a pair of firms for a merger option with synergy δ_k between them. In particular, we want to see how $V_k(\delta_k;\delta_{-k})$ changes with δ_k for a fixed δ_{-k}. We have two interesting findings summarized in Proposition 6.

Proposition 6 (Value of merger option). *In the two-player merger game,*

(i) a merger option that is not exercised has deterrent value if it deters a harmful merger in another tier;

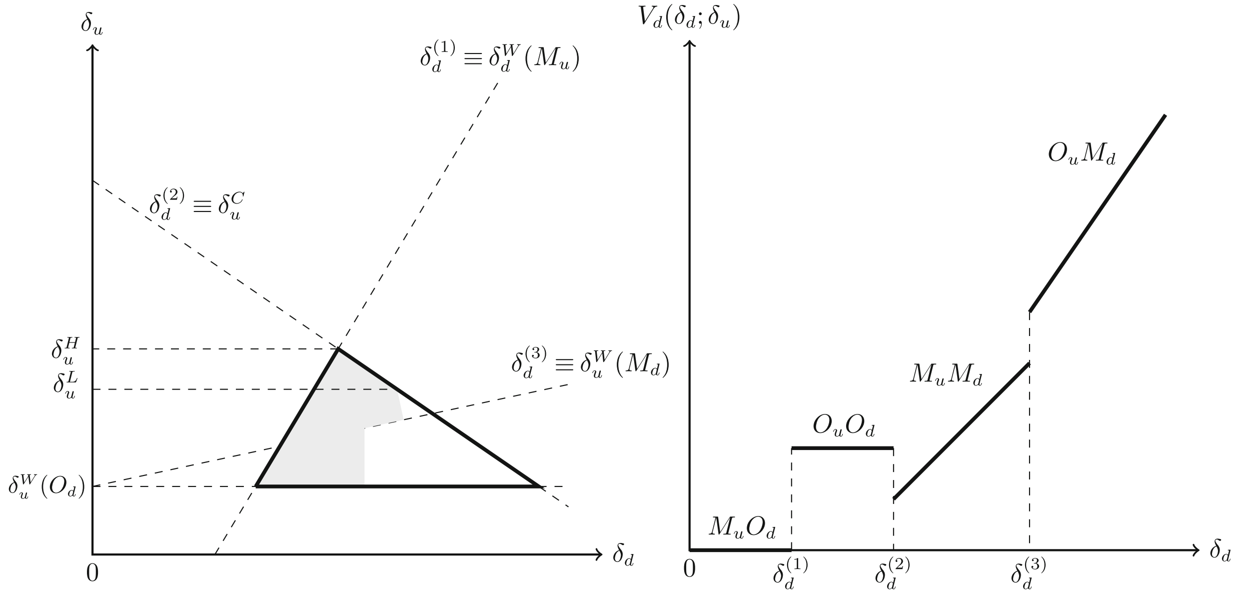
(ii) a merger option with higher synergy may lose the deterrent value and thus the value of the merger option may decease as the synergy increases.

First, a merger option may have deterrent value even though the merger option is not exercised, that is, $V_k(\delta_k;\delta_{-k}) > 0$ though $s_k(\delta_k;\delta_{-k}) = O_k$. In a supply chain, player k's merger option has a strategic deterrent value if it can successfully deter firms in its up- or downstream industry from conducting a merger, as identified in Proposition 4.

Second, the option value may decrease in the merger synergy. Intuition may suggest that the value of a merger option should always (weakly) increase in the merger synergy, but we find that this is not always the case. A merger option's deterrent force becomes weaker as the synergy increases. It implies that one may become worse off if the merger option with a higher synergy cannot deter the other player from merging. Let us take the downstream player for example and focus on an initial state where the downstream player reserves its merger option as a deterrent, that is, the shaded area in Figure 6A. As synergy δ_d increases, the downstream player's gain from his own merger increases but the harm that the merger inflicts on the upstream player weakens, that is, the deterrent credibility becomes higher but the damage to the other player becomes smaller. Thus, as δ_d increases, the downstream player will eventually merge in one of two cases: (i) The downstream player will merge passively since the deterrence method no longer works; (ii) the downstream player will merge proactively while the deterrence method still works. According to previous analysis, given a higher synergy δ_u, it is more difficult for the downstream player to deter the upstream merger. Thus, when the synergy of the upstream merger is high enough, that is, $\delta_u > \delta_u^L$ for some threshold δ_u^L, the first case occurs as δ_d increases starting from somewhere $\delta_d \in (\delta_d^{(1)}, \delta_d^{(2)})$. When $\delta_d > \delta_d^{(2)}$, that is, condition (20) of a deterrent no longer holds, the harm caused by the downstream merger to the upstream player becomes too small and thus the downstream merger option fails to deter the upstream merger. Thus, the downstream player will merge after the upstream player merges and the downstream player becomes worse off at the threshold $\delta_d^{(2)}$. As shown in Figure 6B, given a fixed $\delta_u \in (\delta_u^L, \delta_u^H)$, the equilibrium outcome changes as δ_d increases as follows: For $\delta_d < \delta_d^{(1)}$, neither of the two preventive methods works and the equilibrium outcome is M_uO_d; for $\delta_d \in (\delta_d^{(1)}, \delta_d^{(2)})$, only the deterrence method works and the equilibrium outcome is O_uO_d; for $\delta_d \in (\delta_d^{(2)}, \delta_d^{(3)} \equiv \delta_d^D)$, neither of the two methods works and the equilibrium is M_uM_d; for $\delta_d > \delta_d^{(3)}$, the efficiency effect is high enough such that the downstream merger discourages the upstream merger, that is, only the discouraging method works, and thus the equilibrium outcome is O_uM_d.

The deterrent role of a merger option summarized in Proposition 6 has several interesting managerial implications.

First, a merger option, even an unprofitable one, can have a strategic deterrent value, which equals the harm by another merger if the option is absent. Therefore, even if preserving a merger option is costly, it is worthwhile to take the trouble to maintain it as long as the maintenance cost is smaller than the deterrent value. In practice, for deterrence purpose, the two



(A) (B)

FIGURE 6 Value of downstream merger option $V_d(\delta_d; \delta_u)$ given a fixed $\delta_u \in (\delta_u^L, \delta_u^H)$.

merging firms should sign a *state-contingent* contract which states that they commit to a merger once the other merger is carried out and announce the contract preemptively. The contracting costs, that is, the costs of negotiating, writing, and maintaining a contract between parties, is the actual payment for the merger option. They include the costs of each party verifying the other's information and any legal expenses in the price negotiation process.

Second, when there are two candidate target firms, the acquiring firm should choose the target with a higher synergy in most cases, but there is one exception. For a merger option to qualify as a deterrent to prevent a harmful merger in the other industry, it must meet two conditions simultaneously: credibility and severe damage to firms in the other industry. The latter requires the synergy to be not too high. Thus, firms that have a merger option with higher synergy can be worse off if the merger option can no longer deter the other merger. When a firm has two acquisition targets with different synergies, it may be optimal for the firm to commit to not acquiring the target with higher synergy and to reserve the option to acquire the target with lower synergy purely for deterrence purpose. To enhance the credibility of the commitment, the firm can sign a state-contingent contract with the smaller-synergy target.[13]

Third, when synergy is private information, firms have an incentive to cheat if and only if they can deter a harmful merger by cheating. Richards (2003) argues that in pre-merger talks, as the firms exchange information about customers, suppliers, and technology, they jointly learn the true degree of synergy before the merger is completed. Outsiders do not know the synergy before the merger is completed. We identify an incentive to cheat, while how to credibly cheat is not the focus of our study. Therefore we do not formally model the case with information asymmetry. Richards (2003) gives a potential cheating strategy. During the acquisition negotiation process, when the target firm has some bargaining power, the acquiring firm offers the target firm a share price premium relative to the target firm's existing value in the financial markets, which should reflect the merger surplus and the merger synergy in the absence of information asymmetry. Therefore, under information asymmetry, the merging firms may use the share premium price as a signal to affect outsiders' belief about the synergy of the merger. Richards shows that the acquirer may pay a higher acquisition price to signal a higher synergy than the true level to deter entry. In our setup, either direction is possible: The merging firms may agree on a higher price in order to signal a high synergy such that the deterrent is credible, or they may set a lower price to signal a low synergy such that the deterrent inflicts significant harm on suppliers or customers.

5.3 | Implications for consumer surplus

Thus far, we have discussed two preventive methods in the merger game. That is, two firms in one tier of a supply chain can prevent merger in another tier of the supply chain, if the merger synergy is not large enough, by either merging (Method One) or preserving its merger option (Method Two). In this section, we are going to show that both methods can be beneficial to consumers.

For Method One, recall that a merger tends to discourage mergers in another tier as long as its synergy is high enough. Or equivalently, $\delta^W_{-k}(M_k) > \delta^W_{-k}(O_k)$ if $\delta_k > \delta^*_k$. Now, let's define $\hat{\delta}^*_k = \frac{A_k}{n_k+1}$, which is the threshold that a merger makes

[13]See online appendix for a formal proof of this argument.

no difference to consumers. Then, it is easy to conclude that a merger improves consumer surplus if and only if $\delta_k > \hat{\delta}_k^*$. By comparing δ_k^* with $\hat{\delta}_k^*$, we have $\hat{\delta}_k^* > \delta_k^*$, which indicates discouraging method can be beneficial to consumers only if the merger synergy is large enough.

For Method Two, note that one player can deter another merger by preserving its merger option only if merger synergies in both tiers are moderate. That is, the synergy should be large enough to ensure its own profitability, but can not be too large to be beneficial to firms in other tier as a credible threat. Therefore, for an upstream merger, it can not benefit downstream firms, so it must hurt consumers. Preventing such merger by preserving its merger option obvious benefits consumers. For a downstream merger, it can not benefit upstream firms which also indicates consumers are hurt by the downstream merger. Then, preventing such merger also benefits consumers. In summary, the deterrence method always benefits consumers by preventing mergers.

6 | CONCLUSION

We summarize the findings and their managerial implications. First, our paper is the first study to formally analyze a firm's attitude towards a horizontal merger between its suppliers or customers. The previous literature has consistently found that a horizontal merger in a supply chain is profitable (or beneficial to firms in another tier) if and only if the synergy is above some threshold. Our paper goes one step further by comparing the two thresholds and we find that a firm may have an incentive to do something strategically to prevent a merger in the up- or downstream tier, while it will never have an incentive to precipitate it.

Second, we propose a solution for a firm to prevent such an undesirable merger between its suppliers or customers without relying on government intervention. We suggest that a firm can develop a merger option as an instrument to prevent the harmful target merger. If the synergy of the instrumental merger is high enough, the firm can merge preemptively to discourage the target merger; if the synergies of the instrumental merger and the target merger satisfy such conditions that the instrumental merger can credibly inflict significant harm on the firms involved in the target merger, then the firm can reserve the instrumental merger option to deter the target merger. In some cases, there is a first-mover advantage, that is, the firms involved in each merger want to merge preemptively and the other merger is discouraged. In some other cases, the two mergers are mutual deterrents, and a lose-lose merger war is avoided. These two ways to prevent a target merger explain two phenomena that our model predicts: Firms may conduct an unprofitable merger to discourage another merger; firms will withhold a profitable merger when they are threatened by another merger.

Third, we analyze the value of a merger option and the deterrent role of a merger option has several interesting managerial implications. First, a firm is willing to "pay" for a merger option that it will not carry out. Second, a merger option may lose the deterrent value if the synergy becomes higher. That is, a firm with a low-synergy merger option will not merge but is better off; when the synergy becomes higher, the firm will merge but will become worse off. Therefore, we can infer that for deterrence purpose, a firm may pick a merging partner with a lower synergy among multiple candidates and a firm may signal as a low synergy when synergy is private information before a merger is completed.

Overall, this paper contributes to the literature of mergers in a supply chain by analyzing firms' attitudes toward horizontal mergers between their suppliers or customers and how firms can use their own merger options to strategically prevent mergers between their suppliers or customers. The preventive motive plays a key role in the strategic interaction between mergers across tiers in a supply chain.

Although our results are based on a specific model setup, we think they should be relatively robust. First, most of the interesting phenomena are results of the preventive motive, which exists in most model setups. For example, in a bilateral oligopoly where the downstream buyers have a bargaining power relative to the upstream sellers in negotiating the input price, firms in one tier are usually harmed by mergers in another tier since the merged entity enjoys a larger bargaining power. Provided that a merger can inflict harm on firms in another tier, a merger option can serve as a deterrent. Third, the key finding that it is more demanding for a merger to be beneficial to firms in another tier than to be profitable still holds when we extend the two-tier supply chain model to a multi-tier supply chain model.

According to our research question, we focus on the interaction of horizontal mergers in different tiers of the supply chain. Thus, we assume that there is only one potential two-firm merger in each tier to abstract away from the intra-tier interaction between mergers within the same tier. Though a more realistic setting should allow more than one merger in each tier, including intra-tier and inter-tier interactions would make it difficult to identify the mechanism that how firms can use a merger option to prevent another merger in the up- or downstream tier.

We study the interaction of horizontal mergers between firms in the supply chain by studying a theoretical model. The theoretical modeling allows us to clearly identify the two preventive methods that a firm can do to prevent a horizontal merger in another tier of the supply chain. However, the preventive motive deserves more empirical verification. Moreover, in the real world, there are many factors affecting a firm's merger decision apart from the preventive motives. Future research can conduct surveys on the top managers of firms to better connect the preventive motives to real-world business decisions, and fully identify factors that affect a firm's merger decisions.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

REFERENCES

Ahern, K. R., & Harford, J. (2014). The importance of industry links in merger waves. *The Journal of Finance*, *69*(2), 527–576.

Becker, M. J., & Thomas, S. (2010). The Spillover effects of changes in industry concentration. Unpublished working paper. University of Pittsburgh.

Bhattacharyya, S., & Amrita, N. (2011). Horizontal acquisitions and buying power: A product market analysis. *Journal of Financial Economics*, *99*(1), 97–115.

Cho, S.-H. (2014). Horizontal mergers in multitier decentralized supply chains. *Management Science*, *60*(2), 356–379.

Cho, S.-H., & Wang, X. (2016). Newsvendor mergers. *Management Science*, *63*(2), 298–316.

Fauli-Oller, R. (2000). Takeover waves. *Journal of Economics and Management Strategy*, *9*, 189–210.

Fee, C. E., & Thomas, S. (2004). Sources of gains in horizontal mergers: Evidence from customer, supplier, and rival firms. *Journal of Financial Economics*, *74*, 423–460.

Fridolfsson, S.-O., & Stennek, J. (2005). Why mergers reduce profits and raise share prices–a theory of preemptive mergers. *Journal of the European Economic Association*, *3*, 1083–1104.

Galbraith, J. K. (1952). American Capitalism: The Concept of Countervailing Power.

Ghosh, A., Morita, H., & Wang, C. (2014). Horizontal mergers in the presence of vertical relationships. *Working Paper 3127920, Social Science Electronic Publishing*.

Gorton, G., Kahl, M., & Rosen, R. J. (2009). Eat or Be eaten: A theory of mergers and firm size. *The Journal of Finance*, *64*, 1291–1344.

Gupta, D., & Gerchak, Y. (2002). Quantifying operational synergies in a merger/acquisition. *Management Science*, *48*(4), 517–533.

Horn, H., & Wolinsky, A. (1988). Bilateral monopolies and incentives for merger. *RAND Journal of Economics*, *19*, 408–419.

Inderst, R., & Wey, C. (2003). Bargaining, mergers, and technology choice in bilaterally oligopolistic industries. *RAND Journal of Economics*, *34*, 1–19.

Lommerud, K. E., Straume, O. R., & Sørgard, L. (2005). Downstream merger with upstream market power. *European Economic Review*, *49*, 717–743.

Nilssen, T., & Sørgard, L. (1998). Sequential horizontal mergers. *European Economic Review*, *42*(9), 1683–1702.

O'Brien, D., & Shaffer, G. (2005). Bargaining, bundling, and clout: The portfolio effects of horizontal mergers. *RAND Journal of Economics*, *23*, 299–308.

Oberg, C., & Holtstrom, J. (2006). Are mergers and acquisitions contagious? *Journal of Business Research*, *59*, 1267–1275.

Qiu, L. D., & Zhou, W. (2007). Merger waves: A model of endogenous mergers. *RAND Journal of Economics*, *38*, 214–226.

Richards, D. J. (2003). Mergers and deterrence. *Topics in Economic Analysis & Policy*, *3*(1), 1–18.

Salant, S. W., Switzer, S., & Reynolds, R. J. (1983). Losses from horizontal merger: The effects of an exogenous change in industry structure on Cournot-Nash equilibrium. *Quarterly Journal of Economics*, *98*, 185–199.

Shahrur, H. (2005). Industry structure and horizontal takeovers: Analysis of wealth effects on rivals, suppliers, and corporate customers. *Journal of Financial Economics*, *76*(1), 61–98.

Xiao, Y. (2019). Horizontal mergers under yield uncertainty. *Production and Operations Management*, *29*(1), 24–34.

Yao, Z., & Zhou, W. (2015). Vertical or horizontal: Endogenous merger waves in vertically related industries. *The B.E. Journal of Economic Analysis & Policy*, *15*(3), 1237–1262.

Zhu, J., Boyacib, T., & Ray, S. (2016). Effects of upstream and downstream mergers on supply chain profitability. *European Journal of Operational Research*, *249*, 131–143.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Wu, L., Gu, D., Yao, Z., & Zhou, W. (2023). Creating a merger option to strategically prevent a merger among suppliers or customers. *Naval Research Logistics (NRL)*, 1–15. https://doi.org/10.1002/nav.22148